UNIVERSITY COLLEGE LONDON

EXAMINATION FOR INTERNAL STUDENTS

MODULE CODE : **PHAS0050**

ASSESSMENT PATTERN: **PHAS0050A6PF 001**

MODULE NAME : **PHAS0050 - Climate and Energy**

LEVEL : **Postgraduate (Level 6)**

DATE: : **23/05/2024**

TIME : **14:30**

DURATION : **3 Hours**

This paper is suitable for candidates who attended classes for this module in the following academic year(s):

Year
2023/24

EXAMINATION PAPER CANNOT BE REMOVED FROM THE EXAM HALL. PLACE EXAM PAPER AND ALL COMPLETED SCRIPTS INSIDE THE EXAMINATION ENVELOPE

Hall instructions	
Additional materials	
Standard Calculators	Yes
Non-Standard Calculators	

TURN OVER

Answer ALL THREE questions.

The numbers in square brackets show the provisional allocation of maximum marks per question or part of question.

The materials booklet can be found at the end of the exam paper.

[Part marks]

1. (a) The bandgap in silicon is 1.1 eV. What is the longest photon wavelength that can induce a current in a silicon PV cell? **[1]**

   Consider photons with energies above and below 1.1 eV. What fraction of the incident photon's energy can be captured in a silicon PV in each case? **[2]**

   (b) A silicon PV cell has an open-circuit voltage $V_{\mathrm{OC}} = 0.7$ V. Estimate the maximum power that can be extracted from a solar PV cell producing 1.0 mA of current, making the simplifying assumption that the maximum working voltage $V_m \approx V_{\mathrm{OC}}$. **[2]**

   Estimate the voltage factor, which reflects the ratio of the energy that can be extracted from each electron flowing through the circuit connected to the PV cell, to the energy required to produce the electron-hole pair inside the PV cell. **[2]**

   Briefly describe one other factor, not yet mentioned, which limits the efficiency of PV cells. **[2]**

   (c) Consider a 2 m^2 solar panel in an equatorial region with a peak solar insolation (at midday) equal to 1000 W/m^2. What is the total electrical energy that can be generated over the course of a 24-hour period from this panel, assuming the panel is fixed in an upward pointing direction and has an efficiency of 20%? You can assume that the Sun is directly overhead at midday and neglect the tilt of the Earth's axis of rotation. Furthermore, you can neglect any additional absorption of sunlight in the atmosphere at shallow angles or any other inefficiencies that depend on the angle of incidence of the sunlight. **[4]**

This question continues on the next page ...

... continued from previous page

(d) There is currently a proposal to build a wind and solar plant in Morocco to supply the UK Grid with renewable electricity via an 3800 km HVDC interconnector that has a resistance of 0.01 Ohm/km. Calculate the fractional ohmic energy loss if this HVDC line is used to transmit 1.5 GW of electrical power at a voltage of 500 kV and has a resistance of 0.01 Ohm/km. **[4]**

Identify two challenges to the feasiblity of such a project and suggest how each challenge could be mitigated. **[3]**

[Part marks]

2. (a) Determine the energy released (in units of MeV) by the fission fragments krypton $^{92}_{36}$Kr and barium $^{141}_{56}$Ba assuming that at fission of the parent nucleus they are in contact and that the nuclear radii are given by r = 1.5A$^{\frac{1}{3}}$ fm where A is the mass number, given that $e^2/4\pi\epsilon_0$ = 1.44 MeV fm where e is the proton charge and ϵ_0 is the permittivity of the vacuum. **[2]**

(b) η is defined as the number of fast neutrons emitted per neutron absorbed in a nuclear fuel. Write down an expression for η if the fuel contains x% $^{235}_{92}$U fissile nuclei and $(100 - x)$% $^{238}_{92}$U non-fissile nuclei. **[2]**

Using the data in the table below, evaluate η for a Pressurised Water Reactor (PWR) fuelled with uranium enriched to 4.5% in $^{235}_{92}$U. **[2]**

	$^{233}_{92}$U	$^{235}_{92}$U	$^{239}_{94}$Pu	$^{238}_{92}$U	$^{232}_{90}$Th
σ_f(barns)	469	507	746	-	-
σ_c(barns)	41	87	285	2.4	6.5
ν	2.5	2.4	2.9	-	-
$\eta_{3,5,9}$	2.3	2.05	2.1	-	-

(c) The Westinghouse Small Modular Reactor (SMR) has a thermal output of 800 MW_{th}. Assuming a fuel burn-up of 60 GWd per tonne heavy metal, approximately how many tonnes of UO_2 fuel are required for 2 years of reactor operation between refuelling? **[2]**

This question continues on the next page ...

... continued from previous page

(d) Briefly explain why heat continues to be produced in the core of a reactor even after the fission reaction stops. **[1]**

(e) If the Westinghouse SMR is shut-down in an emergency, approximately what thermal power still needs to be removed by cooling systems? **[1]**

(f) For a Proton-Exchange Membrane (PEM) fuel cell, the maximum available energy is minus the change in the Gibbs free energy:

$$\Delta G = \Delta H - T\Delta S \ ,$$

where the enthalpy change $\Delta H = -285.8\ \mathrm{kJ\,mol^{-1}}$ and the heat dissipated $T\Delta S = 48.7\ \mathrm{kJ\,mol^{-1}}$. Note that these quantities are per mole of hydrogen gas (H_2). Calculate the theoretical maximum efficiency of a PEM fuel cell. **[2]**

(g) Hydrogen gas can be stored in vehicles at a pressure of 350 bars or 35 MPa. Using the free energy given above, and assuming hydrogen behaves as an ideal gas in these conditions, calculate the energy storage density of hydrogen gas at this pressure and a temperature of 20°C. Give your answer in units of $\mathrm{Wh\,litre^{-1}}$. **[4]**

(h) Briefly explain two advantages and two disadvantages of hydrogen powered vehicles compared to battery-electric vehicles. **[4]**

[Part marks]

3. (a) Briefly discuss the evidence for the following assertions, indicating the degree of certainty in each case :

The concentration of CO_2 in the atmosphere is higher than at any time in the last several hundred thousand years. **[2]**

The average surface temperature has increased by approximately 1.2°C since the mid 19th century. **[2]**

The increase in atmospheric CO_2 since the start of the industrial era is largely anthroprogenic in origin. **[2]**

(b) The entropy of N atoms in a volume V at temperature T is given by:

$$S = Nk \ln \left(\frac{V(kT)^{3/2}}{CN} \right) ,$$

where C is a constant and k is Boltzmann's constant. In the above equation, entropic contributions from internal degrees of freedom are neglected. The work of separation is $W = -T \times \Delta S$. Using this result, or otherwise, show that the work required to separate 1 kg of CO_2 from a dilute mixture in which it exists at an initial fraction f is given by:

$$W_{1\text{kg } CO_2} = RT \times \frac{1000}{44} \times \ln(1/f) ,$$

where $R = N_A k = 8.31$ J/mol/K. **[4]**

Calculate $W_{1\text{kg } CO_2}$ for carbon dioxide at an initial concentration of 419 ppm (parts per million by volume) and $T = 287$ K. Express the result in kWh. **[3]**

This question continues on the next page ...

... continued from previous page

(c) Suppose that a solar powered carbon dioxide filtering and sequestration system requires 1 kWh of energy to separate and store 1 kg of CO_2 from the atmosphere. Using the annual solar energy incidence of 1000 kWh/m^2 and a solar power efficiency of 20%, calculate the total area of solar energy collection required to capture 10% of the UK's CO_2 emissions. (The UK's current population is 67 million with an estimated CO_2 emissions of 7 tonnes per year per person.) **[3]**

(d) Briefly discuss some of the advantages the risks of attempting to reduce or reverse global warming through the artificial injection of aerosols into the upper atmosphere. **[4]**

Materials Booklet

Constants

Quantity	Symbol	Value
Boltzmann Constant	k	8.62×10^{-5} eV K^{-1}
Molar Gas Constant	R	8.31 J K^{-1} mol^{-1}
Avogadro Constant	N_A	6.02×10^{23} mol^{-1}
Electron Charge	e	1.602×10^{-19} C
Planck Constant	h	6.626×10^{-34} J s
Molar Volume (ideal gas at STP)	$R \times (273.15\text{ K})/(101,325\text{ Pa})$	22.4×10^{-3} m^3 mol^{-1}
Atomic Mass Unit	amu	931.494 MeV c^{-2}
Acceleration due to Gravity	g	9.81 m s^{-2}
Absolute zero		$-273.15°$C
Density of Water	ρ	1000 kg m^{-3}
Solar Irradiance	S_0	1370 W m^{-2}

Molar Masses

Element	Symbol	Approximate Molar Mass (g mol^{-1})
Hydrogen	H	1.0
Carbon	C	12
Oxygen	O	16
Magnesium	Mg	24
Silicon	Si	28

Standard Integrals

$$\int xe^{-(x/c)^2}\,dx = -\frac{1}{2}c^2e^{-(x/c)^2} + \text{constant}$$

$$\int x^3e^{-(x/c)^2}\,dx = -\frac{1}{2}c^2e^{-(x/c)^2}\left(c^2 + x^2\right) + \text{constant}$$

Basic Thermodynamics

Ideal Gas Law:

$$PV = nRT \ ,$$

where n moles of gas at temperature T occupy volume V and pressure P, and R is the molar gas constant.

First Law of Thermodynamics:

$$Q = \Delta U + W \ ,$$

where the heat energy Q into a system results in a change in internal energy ΔU and the system performing external work W. For a gas or fluid, the change in internal energy is related to the change in temperature through the heat capacity at constant volume: $dU = C_V dT$. The work done by a fluid allowed to expand is $dW = pdV$.

Entropy and Work

For a reversible process, infinitesimal changes in entropy dS are related to heat transfered into the system dQ at temperature T:

$$dS = \frac{dQ}{T} \ .$$

The minimum work required to bring about a change in entropy at fixed temperature is:

$$W_{\text{min}} = -T \times \Delta S \ .$$

Carnot Law:

The maximum efficiency (i.e. the Carnot efficiency) of a heat engine operating between cold and hot reservoirs at temperatures T_C and T_H respectively, is given by:

$$\eta_{\text{max}} = 1 - \frac{T_C}{T_H} \ .$$

Nuclear Power

Neutron cross-sections for fission, capture and scattering are denoted by σ_f, σ_c and σ_s respectively. ν is the average number of fast neutrons produced in a fission event, and η is the average number of fission neutrons produced per thermal neutron absorbed in the fuel. Where subscripts $(3, 5, 9)$ are used, these indicate fissile isotopes ${}^{233}_{92}\mathrm{U}$, ${}^{235}_{92}\mathrm{U}$ and ${}^{239}_{94}\mathrm{Pu}$.

Wind Power

The maximum power that can be delivered by a wind-turbine with cross-sectional area A in up-stream wind-speed u is:

$$P_{\max} = \frac{1}{2}\rho u^3 A \frac{16}{27} ,$$

where ρ is the density of air. The factor of $16/27$ represents the Betz limit on the extracted power.

Hydro and Tidal Power

The gravitation potential energy available when a mass m falls through a height h is given by $E = mgh$, where g is the acceleration due to gravity. The general expression for the power delivered when a fluid of density ρ falls through height h is:

$$P = \eta \rho g h Q ,$$

where Q is the volume flow rate (in $\mathrm{m}^3\,\mathrm{s}^{-1}$) and η is an overall efficiency factor that may need to be applied to make the result more realistic. From these general expressions, the results for dams and pumped storage systems can be derived.

For tidal barrages, a simple estimate for the energy and power that can be delivered using the ebb tide are:

$$E = \frac{\rho g A h^2}{2} \quad \text{and} \quad P = \frac{\rho g A h^2}{2T} ,$$

for a location where the area of water behind the barrage is A and the tidal range is h. T is the tidal period.

For tidal stream power, the maximum power that can be extracted is:

$$P = 0.22 \times \rho g a Q_{\text{max}} \ ,$$

where a is the amplitude of the tidal head difference which drives the flow through the channel, and Q_{max} is the volume flow rate through the channel without any turbines present.

Solar Power

The Planck equation:

$$E = hf \ ,$$

relates the energy of a photon E to the frequency of light f, via Planck's constant h.

The current I_C flowing in a simple circuit incorporating a photocell under illumination and a load resistance R is given by:

$$I_C = I_L - I_S \left(e^{V/V_T} - 1 \right) \ ,$$

where I_L is the current due to the illumination of the photocell. The second term represents the current in the opposite direction flowing through a diode with saturation current I_S and a voltage across it of V. The constant V_T is related to the thermal energies present at temperature T via the expression $V_T = kT/|e|$, where k is Boltzmann's constant and e is the electron charge.

Climate and Climate-Intervention

The average solar power received per square metre of the Earth's surface is:

$$F_S = \frac{S_0}{4} (1 - A) \ ,$$

where S_0 is the solar irradiance at the position of the Earth's orbit around the Sun, and A is the average Albedo of the Earth.